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وكالة البحث العلمي والابتكار

# Smart Expense Management

## A Prototype for Expense Tracking

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## Abstract

This project presents a high-fidelity prototype of a Smart Expense Management application designed to help users track expenses, manage budgets, monitor savings, and improve financial awareness. The prototype was developed using a user-centered design approach, starting with user research, paper prototypes, and culminating in a high-fidelity design in Figma. The system was evaluated using heuristic evaluation, visual design testing, functional usability testing, and survey-based usability testing. The results showed that the prototype supports basic financial management tasks, but improvements are still needed in navigation, visual clarity, icons, and feedback messages.

## Introduction

Managing personal finances can be difficult for young adults because many users do not follow a clear budgeting method or regularly track their expenses. User interviews showed common problems, including overspending at the beginning of the month, impulsive purchases, difficulty saving, and frustration with complex financial apps. Many users also rely on mental estimates or occasional bank checks rather than a structured financial tracking method, making it harder to understand where their money is going. These issues can lead to weak financial awareness, poor spending control, and difficulty maintaining savings goals. This project aims to design a simple Smart Expense Management prototype that helps users record expenses, set budgets, monitor savings, and understand their spending through a clear and easy-to-use interface. The prototype focuses on simplicity, visual clarity, and useful feedback to reduce confusion and support better financial habits. By organizing financial information in one place, the system helps users make more informed decisions and manage their money more easily. Previous studies also show that expense-tracking applications can improve budgeting habits, reduce impulsive spending, and increase users' awareness of their financial behavior [1].

## Methodology

The project followed a user-centered design process, starting with interviews to understand users' financial habits, problems, and expectations. Four participants aged 18–30 were interviewed because they represent the application's main target users. Their responses highlighted common issues, including overspending at the beginning of the month, weak budgeting habits, difficulty saving, and frustration with complex expense-tracking apps. Recurring ideas such as "overspending" and "difficulty saving" helped guide the prototype's main direction. Based on these findings, the main system tasks were identified, including setting budgets, recording expenses, managing savings, monitoring spending behavior, receiving alerts, and viewing monthly summaries. After collecting user needs, the design process began with initial paper prototypes to explore the main screens and system flow. Fig. 1 shows the initial Add Expense sketch used to test the basic form structure. The design was then refined by improving layout organization, titles, buttons, and navigation, as shown in Fig. 2. A final paper prototype was created with additional features, including budget management, savings goals, analytics, profile, notifications, and edit options. Fig. 3 shows the final paper Add Expense prototype before moving to the digital design. Finally, the high-fidelity prototype was created in Figma, where the screens became more realistic and interactive.

The final high-fidelity prototype was evaluated using heuristic evaluation, visual design testing, functional usability testing, and a survey-based usability test. A heuristic evaluation was conducted by another group of students acting as evaluators, using Nielsen's usability principles to identify usability issues. The heuristic evaluation identified issues such as missing Undo options, unclear error messages, and the need for clearer navigation controls. The visual design and functional usability tests were conducted with five participants to evaluate layout, readability, icons, spacing, navigation, and key tasks, including logging in, adding expenses, deleting items, creating savings goals, and editing budgets. The survey-based usability test included 66 participants and measured ease of use, navigation, clarity, and overall user experience. The survey results showed high task completion rates. Fig. 4 shows that 95.5% of participants completed the add-expense task, while Fig. 5 shows that 97% completed the notification-checking task. These results indicate that the main prototype functions were clear and easy to access for most users. Overall, these methods helped identify the prototype's strengths and weaknesses and guided the final improvements.

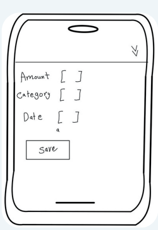


Fig. 1. Initial Add Expense Sketch

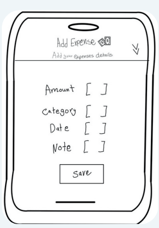


Fig. 2. Refined Add Expense Sketch

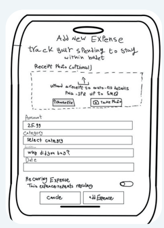


Fig. 3. Final Paper Add Expense Prototype

Fig. 4 shows the task-completion result for adding a new expense.

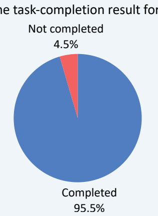


Fig. 4. Add Expense Task Completion

Fig. 5 shows the task-completion results for checking notifications.

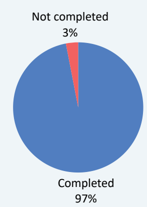


Fig. 5. Notification Task Completion

## Results and Discussion

The results focused on improving the final prototype based on feedback collected during testing. After the heuristic evaluation, visual design test, functional usability test, and survey-based test, several issues were identified in the prototype, such as unclear navigation, crowded layouts, unclear icons, missing confirmation messages, inconsistent feedback, and the lack of an Undo option. Based on these comments, the final design was updated to make the application clearer and easier to use. Navigation was improved to allow users to move between pages more smoothly, and the layout was adjusted to reduce visual clutter and improve readability. Confirmation and feedback messages were also improved, especially for actions such as adding or deleting expenses, so users could better understand what happened after completing a task. The design was also refined by improving spacing, organizing elements more clearly, and making important actions easier to find. Fig. 6 shows the updated Dashboard Overview; Fig. 7 shows the Add Expense Form; Fig. 8 shows the Budget Progress screen; and Fig. 9 shows the Notifications View. This direction is similar to Mint, which also uses dashboards and financial summaries to help users understand their spending. Still, the final prototype focuses more on simple navigation, clear feedback, and reducing user confusion [2]. Overall, the final prototype became more usable and user-friendly because the changes were based directly on user feedback and testing results. These improvements made the Smart Expense Management application clearer, more organized, and easier for users to track expenses, manage budgets, and monitor savings, reducing confusion.

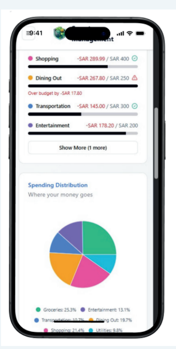


Fig. 6. Dashboard Overview

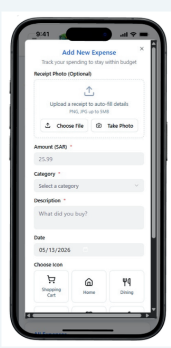


Fig. 7. Add Expense Form

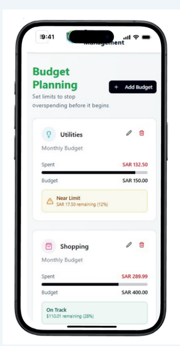


Fig. 8. Budget Progress

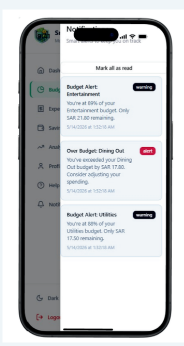

Fig. 9. Notifications View

## Conclusion

The Smart Expense Management prototype was designed to help users track expenses, manage budgets, monitor savings, and understand their spending habits through a simple, clear interface. The project followed a user-centered design process, with user interviews, prototype development, and usability testing used to inform design decisions. The final prototype showed that the application can support basic financial management tasks and improve users' awareness of their spending. Testing also helped identify important usability issues, such as unclear navigation, crowded layouts, unclear icons, and the need for better feedback messages. Based on these findings, the prototype was improved to become clearer, more organized, and easier to use. In future work, the system can be developed into a fully functional mobile application with features such as bank account integration, automatic expense tracking, AI-based spending predictions, and gamification to encourage better long-term financial habits.

## References and Contact

[1] A. P., "The efficiency and adoption of expense tracker applications in enhancing student's financial management," JOIREM, Oct. 2025.
[2] Intuit Inc., "Mint: Personal Finance & Budget Tracker," 2023. [Online]. Available: https://mint.intuit.com

Scan to View Prototype